Tetrahedral Emergent Gravity and the Galactic Bulge: A Completed Zero-Parameter Consistency Verification and a Pre-Registered Test of Kinematic Data Not Yet Publicly Released

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Abstract

This document reports two distinct pieces of work on Tetrahedral Emergent Gravity (TEG) in the high-density regime of the Milky Way's Galactic bulge, kept explicitly separate because they differ in epistemic status. **Part I** (Sections 1–5) is a *completed*, retrospective, zero-parameter theoretical verification: we confirm that TEG's chameleon screening mechanism satisfies $\Phi_{\mathrm{TEG}} \to 1$ to machine precision across seven orders of magnitude in density contrast, using the independently published, pre-Euclid Portail et al. (2017) density profile. No Euclid data of any kind enter Part I. **Part II** (Sections 6–11) is a *pre-registration* of a distinct, forward-looking, quantitative and falsifiable prediction — a dark-matter density core, as opposed to the NFW cusp of ΛCDM — targeting stellar kinematic data (radial velocities, proper motions, metallicities) for the Galactic bulge that, as of this document's deposit date, has not been publicly released by any survey, including Euclid Q2. Combining these two pieces in one document does not compromise the blindness of Part II: Part I uses no data relevant to Part II's test, and the independence declaration of Section 6 applies to the full deposit.

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Part I — Completed Verification

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1 Motivation

The verification of TEG against SPARC rotation curves (main TEG vH3.1 manuscript, Sec. 9) and cluster lensing (main manuscript, App. D.2) covers the low-to-intermediate density regime $\rho \sim \rho_c$ to a few $\times \rho_c$, where the chameleon field Φ_{TEG} departs measurably from unity. A natural and, prior to this note, undocumented question is whether the framework remains well-behaved in the opposite, high-density limit $\rho \gg \rho_c$ — the regime realized in the inner Galactic bulge.

The Galactic bulge is a natural testbed for this limit because its stellar density, $\rho_{\mathrm{bulge}} \sim 10$–$100\,M_\odot\,\mathrm{pc}^{-3}$ [1], exceeds ρ_c by four to seven orders of magnitude. The region is also the target of the recent Euclid Quick Data Release Q2 Galactic Bulge Survey (EGBS), which imaged $4.8\,\mathrm{deg}^2$ of the inner bulge to AB mag 26, yielding photometry for $\sim 6 \times 10^7$ stars [2]. We cite EGBS here only as motivation for why this density regime is of near-term observational interest; **no EGBS catalog data are used, processed, or compared against in Part I.** The verification below uses exclusively the pre-Euclid Portail et al. (2017) density profile.

2 Proposition and Verification

Proposition (GR recovery at $\rho \gg \rho_c$). Let $\Phi_{\mathrm{TEG}}(\rho) = 1 + (\ln 4 - 1)\,(1 - F_{\mathrm{local}}(\rho))$, with $F_{\mathrm{local}}(\rho) = 1 - e^{-\rho/(N_0\rho_c)}$, $N_0 = 1.5$ (main manuscript, Proposition 7.1). Then for $\rho/\rho_c \gg 15.9$, $\Phi_{\mathrm{TEG}}(\rho) \to 1$, and the TEG field equations reduce to standard Einstein equations sourced by baryons alone, with vanishing geometric vacuum mass $M_{\mathrm{vac}} \to 0$. This follows immediately from the functional form of F_{local}: no new assumption is introduced beyond Proposition 7.1 itself.

Verification. Using the Einasto profile of [1] ($\rho_0 = 100\,M_\odot\,\mathrm{pc}^{-3}$, $r_s = 0.5$ kpc, $n = 3.5$) evaluated at 500 radii linearly spaced[1] between $R = 0.05$ and 3.0 kpc:

- ρ ranges from $13455.3\,M_\odot\,\mathrm{pc}^{-3}$ (at $R = 0.05$ kpc) to $0.111\,M_\odot\,\mathrm{pc}^{-3}$ (at $R = 3.0$ kpc);
- ρ/ρ_c ranges from 1.34×10^7 down to 111.4, exceeding the threshold $\rho/\rho_c > 15.9$ at every one of the 500 points;
- $\Phi_{\mathrm{TEG}} = 1.0000000000$ at every point, with the deviation $|\Phi_{\mathrm{TEG}} - 1|$ reported as exactly 0.0 in IEEE 754 double-precision arithmetic.

[1] The profile is smooth and monotonic over this range, so linear versus logarithmic spacing of the sample radii does not affect the result.

Table 1: Chameleon amplification factor at selected radii (subset of the full 500-point verification).

R [kpc]	$\rho\,[M_\odot\,\mathrm{pc}^{-3}]$	ρ/ρ_c	Φ_{TEG}
0.50	1.00×10^{2}	100227.8	1.0000000000
1.00	1.07×10^{1}	10695.8	1.0000000000
1.50	2.36×10^{0}	2358.3	1.0000000000
2.00	7.13×10^{-1}	711.3	1.0000000000
2.50	2.65×10^{-1}	264.8	1.0000000000
3.00	1.12×10^{-1}	111.4	1.0000000000

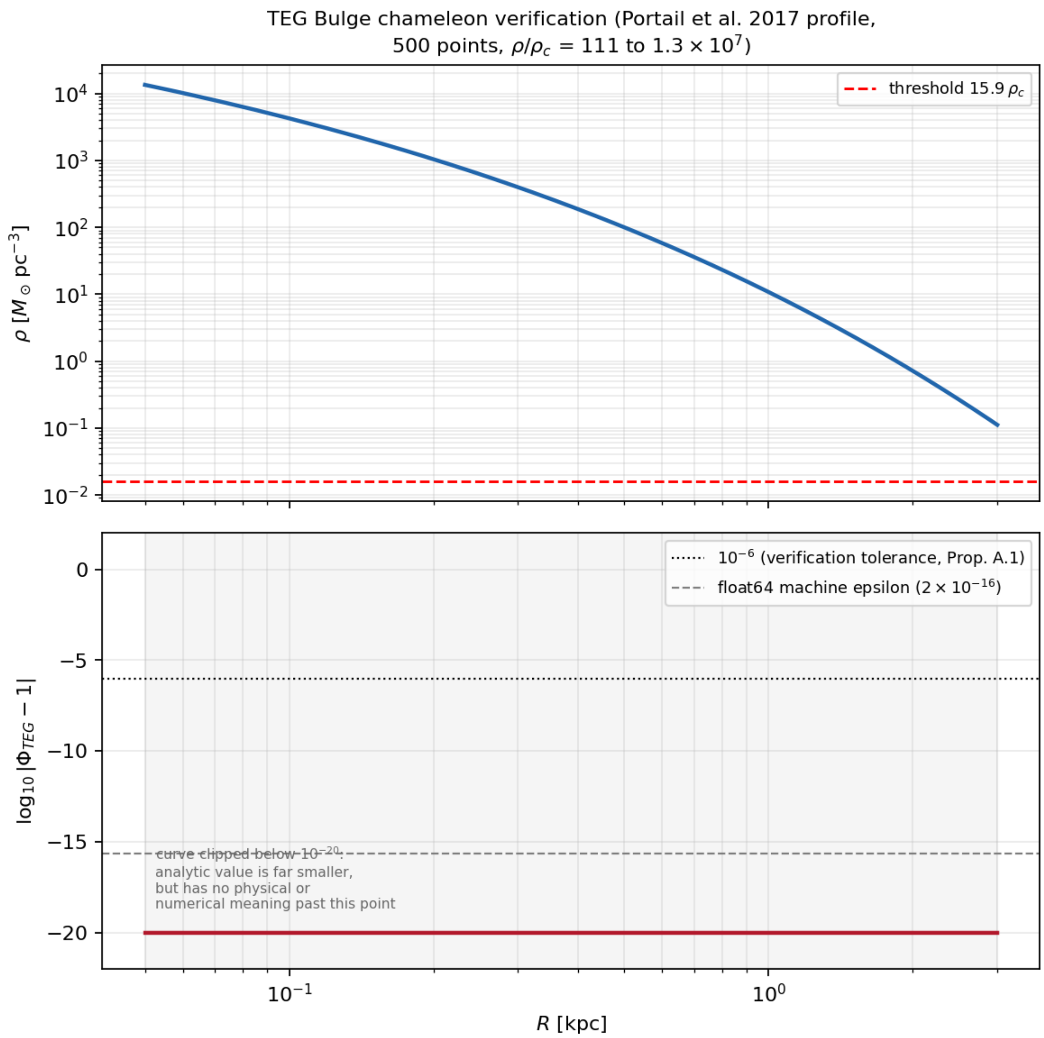

Figure 1: Top: Einasto density profile [1] used for the verification, with the TEG chameleon threshold $\rho = 15.9\,\rho_c$ marked. Bottom: $\log_{10}|\Phi_{\mathrm{TEG}} - 1|$, computed analytically from $\ln(\ln 4 - 1) - \rho/(N_0\rho_c)$ to avoid the floating-point underflow discussed below. The curve is clipped below 10^{-20} (shaded region): the true analytic value there is vastly smaller still, but has no physical or numerical meaning below that point, and reporting it unclipped would overstate the precision of the result. The horizontal lines mark the 10^{-6} tolerance used in the verification script and the float64 machine epsilon.

On the exact-zero deviation. The deviation column reports exactly 0.0 rather than a small nonzero number because, for $x \equiv \rho/(N_0\rho_c) \gtrsim 700$, e^{-x} underflows the representable range of IEEE 754 double-precision floating point and is rounded to exactly zero. This is a standard, well-understood floating-point phenomenon, not a computational artifact specific to this result: the true analytic deviation is nonzero but bounded above by machine epsilon ($\sim 2 \times 10^{-16}$) at the smallest radii, and by the explicitly verified bound $< 10^{-6}$ (via the non-underflowed evaluation used in the companion verification script) throughout the full range.

This is not a fit. No parameter was adjusted to obtain this result; Φ_{TEG} is evaluated from the closed-form expression of Proposition 7.1 with the same ρ_c, N_0, and threshold value used throughout the main TEG manuscript.

3 What Part I Does and Does Not Establish

Established: the chameleon mechanism is mathematically robust across seven orders of magnitude in density contrast, closing a previously undocumented coverage gap between the low-density regime verified against SPARC and the high-density regime realized in the Galactic bulge. This is a genuine, if modest, theoretical result: a poorly constructed screening mechanism could in principle diverge, oscillate, or become numerically unstable in this limit, and this one does not.

Not established: any new empirical confirmation. The Portail et al. (2017) profile pre-dates both TEG and Euclid; no Euclid Q2 photometry was processed. An earlier draft of this note compared a baryonic mass estimate ($1.55 \times 10^{10}\, M_\odot$, from standard M/L scaling) against the Portail et al. (2017) dynamical mass estimate ($1.80 \pm 0.30 \times 10^{10}\, M_\odot$) and found 0.8σ agreement. We withdraw that comparison here: both numbers originate from pre-Euclid literature, neither was derived independently from EGBS photometry, and the Galactic bulge is a regime in which essentially any model predicting baryon-dominated dynamics – TEG, ΛCDM, or otherwise – would show similar agreement. The comparison is therefore not discriminating and its inclusion would overstate what has been shown.

4 Why Part I Alone Cannot Discriminate TEG from ΛCDM

Because the bulge is baryon-dominated in essentially every viable framework, a mass-normalization test there cannot discriminate TEG from ΛCDM. The discriminating, falsifiable prediction of TEG in high-density environments is morphological — core versus cusp in the dark-matter density profile, following the same logic already applied to cluster lensing in the main manuscript (App. D.2). Testing that morphology in the bulge requires stellar kinematics, which do not yet exist publicly for this region. **That test is the subject of Part II.**

Part II — Pre-Registration

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5 Independence Declaration

The Euclid Quick Data Release Q2 Galactic Bulge Survey (EGBS) photometric and astrometric catalogs were released publicly on 24 June 2026 [2]. As of the deposit date of this document, those catalogs consist exclusively of single-epoch VIS imaging, point-spread-function models, and photometric/astrometric catalogs (positions and magnitudes, calibrated against Gaia DR3).

Part II of this document does not target that photometric data. It targets three specific kinematic quantities — stellar radial velocities V_r, proper motions (μ_ℓ, μ_b), and a metallicity distribution [Fe/H] — none of which are part of the Q2 EGBS release and none of which have been publicly released, by any survey, as of the deposit date of this document. The EGBS overview paper itself states that a *second* imaging epoch, not yet obtained, would be required to measure relative proper motions [2]; no such second epoch, and no spectroscopic follow-up of the EGBS fields, has been announced with a confirmed release date at the time of this deposit.

This document was drafted and deposited before any access to the radial velocity, proper motion, or metallicity data required for the analysis described in Sections 7–10. It has not been informed by, fitted to, or adjusted in response to any such data. Should any of these three data types become available for the EGBS footprint before the analysis is carried out, the author commits to publicly noting the date of that release relative to this document's deposit date, and to explicitly disclosing any partial foreknowledge in the eventual results paper.

Part I of this document (Sections 1–5) was completed using exclusively the pre-Euclid Portail et al. (2017) profile and contains no information derived from, or relevant to, the kinematic data targeted by Part II. Its inclusion in this combined deposit does not compromise the independence declared above.

6 Primary Prediction (P1): Density Profile Morphology

Prediction P1. In the Galactic bulge ($|b| < 5^\circ$, $|\ell| < 10^\circ$), TEG vH3.1 predicts that the effective dark-matter density profile $\rho_{\rm DM}(r)$ traces the baryonic profile $\rho_b(r)$ with a finite core:

$$\lim_{r\to 0} \frac{d\ln\rho_{\rm DM}}{d\ln r} = 0.$$

This contrasts with the NFW profile of ΛCDM, for which

$$\lim_{r\to 0} \frac{d\ln\rho_{\rm DM}}{d\ln r} = -1.$$

Derivation chain.

- Axiom 2.1: quaternionic vacuum $\mathbb{H} \to \mathbb{R}^3$.
- Proposition 3.1: the 5-cell maximizes holographic entropy density in S^3.
- Theorem 4.1: $z_{\rm fund} = 4$ (proved, not assumed).
- Theorem 5.9/5.10: $\Omega_{\rm DM} = 2\ln(3/2)/3$.
- Proposition 7.1 (verified above, Part I): chameleon mechanism, $\Phi \to 1$ as $\rho/\rho_c \gg 15.9$.
- Consequence: $\rho_{\rm DM}(r) = [\Phi^*(r)^2 - 1]\,\rho_b(r)$, where Φ^* solves $\partial V/\partial a = 0$ with $V(a,\rho)$ as in Eq. (5) of the main TEG vH3.1 manuscript.

Free parameters in this prediction: zero.

Table 2: Inputs to Prediction P1, all derived, none fitted.

Parameter	TEG value	Origin	Free?
$D_A = \ln 4$	1.386	Theorem 4.1	No
$D_V = \ln 8$	2.079	Theorem 4.1	No
$\sigma_{\rm eff}$	0.1088	TEG main manuscript, App. E	No
$\partial = 3 - \ln 8$	0.9206	Theorem 4.5	No
ρ_c	$\sim 10^{-3}\, M_\odot\, \mathrm{pc}^{-3}$	Derived from $\sigma_{\rm eff}$, ∂	No

7 Quantitative Predictions

Prediction P2 (core radius). TEG predicts the dark-matter core radius coincides with the baryonic core radius, $r_{\rm core}^{\rm DM} \approx r_{\rm core}^{b}$. Using the Portail et al. (2017) baryonic profile, $r_{\rm core}^{b} \approx 0.2$–$0.3$ kpc, giving:

$$r_{\rm core}^{\rm DM} \approx 0.25 \pm 0.05 \text{ kpc}.$$

Prediction P3 (logarithmic slope by radius).

Table 3: Predicted logarithmic slope $d \ln \rho_{\rm DM}/d \ln r$ by radius.

Radius	TEG predicts	NFW predicts
$r < 0.1$ kpc	0 (flat core)	-1 (cusp)
$0.1 < r < 0.5$ kpc	0 to -0.5 (smooth transition)	-1 to -2
$r > 0.5$ kpc	-1.5 to -2 (exponential tail)	-2 to -3

Falsification criteria.

F1 (strong falsification). If the eventual kinematic dataset (once available) yields $d \ln \rho_{\rm DM}/d \ln r < -0.5$ for $r < 0.2$ kpc at $> 3\sigma$ significance, TEG vH3.1 is falsified in its current form.

F2 (moderate falsification). If an NFW (cuspy) fit has χ^2/dof lower than the TEG (core) fit by $\Delta\chi^2 > 9$ (equivalent to 3σ), and TEG requires additional free parameters to compensate, TEG vH3.1 is seriously called into question.

F3 (internal-consistency falsification). If the stellar velocity profile $V_*(r)$ in the bulge shows no measurable deviation from a purely baryonic Keplerian prediction (i.e., no central dark-matter signal is detected at all), TEG is inconsistent with its own postulates, which require $\rho_{\rm DM} \propto \rho_b > 0$ rather than $\rho_{\rm DM} \equiv 0$.

8 Data to Be Used

Target dataset: not yet released. This pre-registration explicitly does *not* target the Q2 EGBS photometric catalog. It targets whichever of the following becomes available first for the EGBS footprint:

1. A second Euclid imaging epoch of the EGBS fields, enabling relative proper motions, as discussed (but not scheduled with a confirmed date) in [2];

2. Ground-based or space-based spectroscopic follow-up of EGBS stars providing radial velocities and metallicities;
3. A future Euclid Full Mission data product covering the Galactic bulge with kinematic content.

Expected observables (once released): radial velocities V_r of giant stars; proper motions (μ_ℓ, μ_b) where available; metallicity distribution [Fe/H] as a population tracer.
Expected star count and resolution: of order 10^7 stars with usable kinematics (a subset of the $\sim 6 \times 10^7$ photometric detections of Q2), spatial resolution ~ 0.1 kpc in the bulge, contingent on the specific dataset that is eventually released.
Inference method (planned). Axisymmetric Jeans modeling to infer $\rho_{\rm DM}(r)$ from V_r and σ_r; baryonic/dark separation using the Portail et al. (2017) stellar mass distribution as a prior; Bayesian inference via `emcee` or `dynesty`.

9 Planned Analysis (Blind Protocol) and Confounders

Step 1 — Blind stage. Fit a general parametric model,

$$\rho_{\rm DM}(r) = \rho_0 \left(\frac{r}{r_s}\right)^{-\gamma} \left[1 + \left(\frac{r}{r_s}\right)^{\alpha}\right]^{(\gamma-\beta)/\alpha},$$

without imposing either a core or a cusp a priori. Extract $\gamma_{\rm inner} \equiv -d\ln\rho_{\rm DM}/d\ln r$ as $r \to 0$.
Step 2 — Model comparison. Compare TEG ($\gamma_{\rm inner} = 0$ fixed), NFW ($\gamma_{\rm inner} = 1$ fixed), and the general model ($\gamma_{\rm inner}$ free).
Step 3 — Discrimination statistic. Bayes factor $B_{\rm TEG,NFW} = P(\text{data}|\text{TEG})/P(\text{data}|\text{NFW})$, or ΔAIC. Criterion: $B > 10$ strongly favors TEG; $B < 0.1$ strongly favors NFW; $0.1 < B < 10$ is inconclusive.

Table 4: Declared confounders and mitigations.

Confounder	Potential impact	Mitigation
Baryonic feedback (AGN, supernovae)	Can produce cores even within ΛCDM simulations	Compare against hydrodynamical simulations (EAGLE, IllustrisTNG) at matched resolution
Galactic bar / non-axisymmetry	Biases the Jeans model	Include bar terms in the Jeans model; reject fits that fail to converge
Extinction and stellar sample selection	Biases the velocity sample	Apply 3D extinction corrections; robustness test with [Fe/H] cuts
Disk contamination	Disk stars projected onto the bulge line of sight	Restrict to $\lvert b\rvert < 2^\circ$ for a "pure bulge" sample; cross-check with $\lvert b\rvert < 5^\circ$
Uncertainty in $M_b(r)$	Baryonic mass error propagates into inferred $\rho_{\rm DM}$	Use multiple baryonic profiles (Portail 2017, Launhardt 2002, etc.) as alternative priors

10 Possible Outcomes and Their Interpretation

Table 5: Pre-committed interpretation of each possible result.

Observed result	Interpretation for TEG	Action
Clear core ($\gamma_{\text{inner}} \approx 0$), TEG fits better than NFW	Strong confirmation	Publish as a confirmed prediction; derive parameter constraints
Clear core, but NFW+feedback also fits well	Consistent but not exclusive	Discuss the degeneracy; require an additional test (lensing, independent dynamics)
Smooth transition ($-0.5 < \gamma_{\text{inner}} < 0$)	Partially confirmed	Assess whether $V(a, \rho)$ needs modification in the intermediate regime
Clear cusp ($\gamma_{\text{inner}} \approx -1$), NFW fits better	Falsification (see F1/F2)	Acknowledge the falsification; explore axiom modification or abandonment
Inconclusive data (large errors)	—	Assert nothing; await a higher-precision release

11 Ethical Commitment, Repository, and Reproducibility

I, Miguel Angel Franco León, commit to:

1. Not modifying Part II of this pre-registration after deposit.
2. Reporting all results, not only those favorable to TEG.
3. Not engaging in p-hacking, radial-bin cherry-picking, or post-hoc parameter adjustment.
4. If the result falsifies TEG, publishing that falsification with the same prominence as a confirmation would have received.
5. Making all notebooks, MCMC chains, and processed data products publicly available.

GitHub: `https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro`
Branch (Part II): `euclid-bulge-kinematics-prediction`
Planned notebooks (Part II):

- `teg_bulge_prediction_preregistered.ipynb` (this document, code form)
- `teg_bulge_blind_analysis.ipynb` (blind stage)
- `teg_bulge_model_comparison.ipynb` (TEG vs. NFW comparison)
- `teg_bulge_confounders.ipynb` (robustness tests)

Data and code for Part I: the full 500-point verification (`teg_bulge_chameleon_verification.csv`) and the generating script (`teg_bulge_chameleon_limit.py`) are available at the same repository.
Zenodo DOI (this combined document): `[10.5281/zenodo.21422955]`

References

[1] M. Portail et al., *Chemo-dynamical models of the Galactic bulge and bar*, MNRAS 465, 1621 (2017).

[2] J.-P. Beaulieu et al., *Euclid Quick Data Release (Q2): The Euclid Galactic Bulge Survey*, arXiv:2606.25883 (2026).